Raj Narayan Singh

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Profile

Computer Science student at KIIT University with hands-on experience in AI systems engineering, workflow automation, LLM integration, and scalable backend development. Currently developing production-oriented Agentic AI systems involving multi-step execution pipelines, workflow orchestration, conversational AI, and real-time backend automation using FastAPI, Python, and Generative AI technologies. Strong foundation in AI infrastructure, modular backend architecture, predictive ML systems, and data-driven engineering workflows.

Technical Skills

Programming & Data: Python (Pandas, NumPy, Scikit-learn), SQL (Joins, Subqueries, CTEs), Java
AI, LLMs & Agentic Systems: Machine Learning, NLP, Generative AI, LLM Integration, Prompt Engineering, Agentic AI, Workflow Orchestration, LangGraph, Sentence Transformers, FAISS, Structured Output Handling, State Management
AI Automation & Backend: FastAPI, REST APIs, Webhooks, Conversational AI, Voice AI Workflows, Workflow Automation, Backend Integration
Data Analytics & Visualization: Power BI (DAX, Power Query), Excel, Exploratory Data Analysis (EDA), KPI Dashboards, Statistical Analysis, Plotly, Matplotlib
Deployment & Tools: Streamlit, Git/GitHub, Model Deployment, Joblib

Professional Experience

AI Engineering Intern — 05/2026 – Present
Hamath OPC Pvt Ltd — *Remote*

- Developing AI-powered workflow automation and Agentic AI systems focused on multi-step execution pipelines, backend orchestration, and scalable AI integrations using FastAPI and LLM-based architectures.
- Building production-oriented AI workflow systems integrating conversational AI, structured execution pipelines, external APIs/tools, workflow state management, and intelligent response handling.
- Designed modular backend services for AI orchestration, request validation, automation pipelines, and real-time workflow processing with scalable system architecture principles.
- Exploring Agentic AI concepts including orchestration logic, autonomous workflow execution, structured output generation, and backend-driven AI automation systems.

Business Analyst Intern — 06/2025 – 12/2025
Labmentix Pvt Ltd — *Remote*

- Engineered automated Power BI dashboards and KPI reporting pipelines using Python and SQL, reducing manual reporting effort by ∼40%.
- Performed EDA on multi-domain business datasets, surfacing actionable operational and product insights.
- Leveraged Generative AI and Groq API for SQL drafting, EDA documentation, and insight summarization, reducing analytics turnaround time by ∼30%.

Projects

AI Startup Evaluator | GitHub Repo

- Architected a multi-component AI evaluation system: NLP-based keyword extraction and topic classification via Sentence Transformers to derive structured feature scores across market size, innovation, scalability, competition, and feasibility dimensions.
- Integrated Groq API (LLM) to generate competitor insights, SWOT analysis, risk assessments, and market opportunity summaries; implemented FAISS vector database with sentence-transformer embeddings to compare startup ideas against similar companies and measure market saturation.
- Deployed full-stack application with Streamlit frontend and FastAPI backend featuring Plotly/Altair visualizations — success probability gauge, industry comparison charts, and innovation score — enabling startup viability evaluation in under 60 seconds.

Aircraft Engine Predictive Maintenance System | GitHub Repo

- Built an end-to-end ML pipeline on the NASA CMAPSS Turbofan dataset; computed RUL labels per engine cycle and engineered rolling statistics, lag features, and cycle-progression features across 20+ sensors to capture degradation trends.
- Benchmarked 5 algorithms (Linear Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost) with Grid Search / Randomized Search tuning; Random Forest and Gradient Boosting achieved cross-validated R^2 of 0.87 and RMSE ∼18 cycles, outperforming baseline by 34%.
- Deployed an interactive Streamlit dashboard (Joblib-serialized model) for live sensor input, visualizing predicted RUL, degradation curves, and sensor patterns — delivering ∼15–20 cycle advance warnings for proactive maintenance scheduling.

Education

Kalinga Institute of Industrial Technology (KIIT University) — Bhubaneswar
B.Tech – Computer Science *GPA: 8.23 / 10.0* — *09/2022 – 05/2026*